

Hands-On-Learning: Create a Patch Management SOP

Learning Objectives

By the end of this activity, learners will be able to:

- Structure a formal Standard Operating Procedure (SOP) for a critical security process.
 - Use an AI assistant (Microsoft Word Copilot) to draft technical and procedural content.
 - Define clear timelines, roles, and escalation paths for a patch management lifecycle aligned with ISO and NIST principles.
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Description

This lab places the learner in the role of an IT Operations Lead responsible for formalizing a critical security function: patch management. Learners will design a Standard Operating Procedure (SOP) that outlines the end-to-end process for identifying, testing, and deploying patches. The activity focuses on creating clear, repeatable, and auditable documentation, a core component of implementing and maintaining a robust governance framework.

Structure of the Activity

Title: Create a Patch Management SOP

Tool used:

Microsoft Word with Copilot

Scenario

You are the IT Operations Lead at "LogiCore," a logistics company that relies on a complex mix of on-premise servers and cloud infrastructure. During a recent internal review, auditors found that the company's patch management process was ad-hoc, inconsistent, and poorly documented, posing a significant risk. Your manager has tasked you with creating a formal SOP that aligns with best practices like those in NIST SP 800-40 to ensure vulnerabilities are addressed in a timely and structured manner.

Objective

To build a patch management SOP using Microsoft Word Copilot that integrates clear timelines, escalation paths, and documentation requirements to ensure a timely and auditable response to vulnerabilities.



Instructions for learners

1 Structure the SOP:

Open a new Microsoft Word document.

Create the main headings for your SOP:

1.0 Purpose

2.0 Scope

3.0 Roles and Responsibilities

4.0 Procedure (with sub-headings: 4.1 Identification, 4.2 Prioritization, 4.3 Testing, 4.4 Deployment, 4.5 Verification)

5.0 Timelines and SLAs

6.0 Escalation Path

2 Draft Content with Copilot:

Use Copilot to generate content for each section.

Example Prompt for section 5.0: "Draft a section for a patch management SOP titled 'Timelines and SLAs'. Include a table with patch timelines based on vulnerability severity (Critical, High, Medium, Low) according to NIST SP 800-40."

Use another prompt for the escalation path.

Example Prompt for section 6.0: "Draft an escalation path for patches that fail deployment. Start with notification to the IT Ops Lead, then escalate to the IT Manager after 24 hours, and finally to the CISO after 48 hours."

Review and lightly edit the AI-generated text to ensure it is clear and fits the LogiCore scenario.

3 Finalize the Document:

Under "Roles and Responsibilities," explicitly define the duties for the "IT Operations Lead" and "System Administrator."

Ensure the document has a professional and organized appearance.

Organize Information Following This Structure for Peer Review

Title Page: Project title, your name, date.

SOP Document: Submit the completed Patch Management SOP as a PDF document (1-2 pages).

AI Prompt Used: In your submission post or a separate section, share the most effective prompt you used with Word Copilot and briefly explain why it worked well.

Audit-Readiness Justification: In one paragraph, explain how this SOP helps LogiCore demonstrate control effectiveness to an auditor.

Instructions for Documenting and Sharing the Project for Peer Review

Document the Project



Ensure your SOP is complete and well-formatted. Proofread for clarity and accuracy.

Share the Project



Save your SOP in PDF format. Upload your document to the “Peer Review” area in the course shell. Provide a brief description of your project in the submission post.



Peer Review

Review the projects submitted by your peers and provide constructive feedback on their work.

This activity gives learners a chance to apply the lesson by creating a formal, auditable document that operationalizes governance policy, turning abstract requirements into a concrete, repeatable process.